

ELEC3609 Individual Task 5

Understand Authentication, Sessions, and Cookies

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Task 1 - Django Authentication System

1. What is the difference between authentication and authorisation in Django?

Authentication checks who the user is, such as by checking a username and password.

Authorisation checks what that logged-in user is allowed to access or change. In Django, authentication backends handle the login check, while permissions, groups and checks in views are used for authorisation.

2. How does Django remember logged-in users between requests?

When a user logs in, Django saves the user's ID and authentication backend in the session. The browser keeps a session cookie and sends it with later requests, which lets Django load the session and set `request.user`. With the normal database session backend, the actual session data stays on the server rather than inside the cookie.

3. What are cookies used for, and are all cookies safe?

Cookies are small pieces of data stored in the browser and are commonly used for login sessions, preferences, shopping carts and analytics. They are not always safe because a stolen session cookie may let someone access an account, and tracking cookies can affect privacy. Important cookies should use HTTPS and settings such as `Secure`, `HttpOnly` and `SameSite`, and sensitive information should not be stored directly in them.

4. Why might a project use a custom user model or authentication backend?

A custom user model is useful if the project needs something different from Django's default user, such as using email for login or storing extra required account details. A custom authentication backend can also connect Django to SSO, LDAP or another login service, or apply a different way of checking credentials and permissions.

Task 1 - Continued

5. How should user fields be updated safely after development has started?

The project should not replace AUTH_USER_MODEL just to add or change a field. It is safer to edit the existing custom user model, or add a separate profile model, and then create and review a Django migration. For a new required field, I would first allow it to be blank or give it a safe default, fill the old records with a data migration, and then add the final required constraint after testing on a backup.

Task 2 - Project Progress Check-in

1. Current role

Both frontend and backend. I am mainly focusing on the backend structure, authentication and security, and connecting the backend to the frontend.

2. Features I am responsible for

The main areas I am working on are the registration and login flow, email OTP and session security, and the project and task features such as invitations, task assignments, status updates and comments.

3. Progress update

I have finished the main authentication flow and connected the Django API to the React workspace. The project and task workflows are working locally, and I have also added automated tests and checked the backend and frontend builds.

4. Estimated completion

I think the project has reached about 90% completion locally. The remaining work should be completed by the end of Week 7 after the team finishes the AWS deployment and collects the final submission evidence.

5. Team feedback or concerns

There is no major teamwork problem at the moment. My main concern is making sure the AWS deployment, video, contract and Canvas evidence are divided clearly, because these remaining tasks need contributions from the whole team.